

Question

Dissolving $S \text{ mol } Ag_2Cr_2O_7$ in 1 L of pure water results in saturation, meaning there is no precipitate, and the solution volume remains unchanged. The following data are known (activity coefficients are not considered):

$$K_{sp}(Ag_2Cr_2O_7) = 1.99 \times 10^{-7}$$

$$K_{sp}(Ag_2CrO_4) = 1.12 \times 10^{-12}$$

$$K_{sp}(AgOH) = 2.01 \times 10^{-8}$$

$$K_a(HCrO_4^-) = 3.23 \times 10^{-7}$$

$$2HCrO_4^- = Cr_2O_7^{2-} + H_2O, K_f = 1.14 \times 10^2$$

H_2CrO_4 is considered a strong acid upon its first ionization and a weak acid upon its second ionization; $H_2Cr_2O_7$ is considered a diprotic strong acid.

Select the correct option(s) below.

A. All other options are incorrect

B. at equilibrium, $4.20 \leq pH \leq 4.50$

C. At equilibrium, $0.95 \leq [Ag^+]^2[Cr_2O_7^{2-}]/K_{sp}(Ag_2CrO_4) \leq 1.05$ (Note: Not a typo, the denominator is indeed Ag_2CrO_4)

D. The mass $m(g)$ of $Ag_2Cr_2O_7$ added at exact saturation satisfies $5.5 \times 10^{-2} \leq m \leq 5.8 \times 10^{-2}$

E. Adding $2S \text{ mol}$ of $Ag_2Cr_2O_7$ to 1 L of pure water, at equilibrium the precipitate contains both Ag_2CrO_4 and $Ag_2Cr_2O_7$.

F. Adding $S \text{ mol } Ag_2CrO_4$ to 1 L of pure water, it can still be completely dissolved.

Answer

Correct Answer: **A**

Detailed Explanation

At equilibrium, the solution is obviously weakly acidic. We can first not consider $AgOH$ saturation. Then either $Ag_2Cr_2O_7$ or Ag_2CrO_4 is saturated, that is, one of the following two equations holds as an equality, and the other equation holds as a less than:

$$[Ag^+]^2[Cr_2O_7^{2-}] = K_{sp}(Ag_2Cr_2O_7) \text{ (Equation 1)}$$

$$[Ag^+]^2[CrO_4^{2-}] = K_{sp}(Ag_2CrO_4) \text{ (Equation 2)}$$

CHECKPOINT

1 PTS

One of $[Ag^+]^2[Cr_2O_7^{2-}] = K_{sp}(Ag_2Cr_2O_7)$ and $[Ag^+]^2[CrO_4^{2-}] = K_{sp}(Ag_2CrO_4)$ holds, and the other equation holds as a less than

When completely dissolved, according to the conservation of atoms $[Ag] = [Cr]$, we have

$$[Ag^+] = 2[Cr_2O_7^{2-}] + [HCrO_4^-] + [CrO_4^{2-}] \text{ (Equation 3)}$$

CHECKPOINT

1 PTS

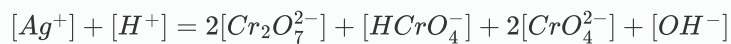
$$[Ag^+] = 2[Cr_2O_7^{2-}] + [HCrO_4^-] + [CrO_4^{2-}]$$

According to charge conservation, we have

$$[Ag^+] + [H^+] = 2[Cr_2O_7^{2-}] + [HCrO_4^-] + 2[CrO_4^{2-}] + [OH^-] \text{ (Equation 4, } [OH^-] \text{ can also be ignored)}$$

CHECKPOINT

1 PTS

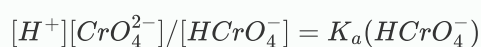


According to K_a and K_f , we have

$$[H^+][CrO_4^{2-}]/[HCrO_4^-] = K_a(HCrO_4^-) \text{ (Equation 5)}$$

CHECKPOINT

0.5 PTS



$$[Cr_2O_7^{2-}]/[HCrO_4^-]^2 = K_f \text{ (Equation 6)}$$

CHECKPOINT

0.5 PTS



If $Ag_2Cr_2O_7$ is saturated, then take Equations 1 and 3-6, and solve them simultaneously (units are all mol/L , the same below)

$$[Ag^+] = 9.25 \times 10^{-3}, [H^+] = 3.78 \times 10^{-5}, [Cr_2O_7^{2-}] = 2.34 \times 10^{-3}, [HCrO_4^-] = 4.53 \times 10^{-3}, [CrO_4^{2-}] = 3.88 \times 10^{-5}$$

But at this time $[Ag^+]^2[CrO_4^{2-}] > K_{sp}(Ag_2CrO_4)$, discard it.

CHECKPOINT

3 PTS

The assumption that $Ag_2Cr_2O_7$ is just saturated is not reasonable.

If Ag_2CrO_4 is saturated, then take Equations 2-6 and solve them simultaneously

$$[Ag^+] = 3.36 \times 10^{-4}, [H^+] = 9.92 \times 10^{-6}, [Cr_2O_7^{2-}] = 1.06 \times 10^{-5}, [HCrO_4^-] = 3.05 \times 10^{-4}, [CrO_4^{2-}] = 9.94 \times 10^{-6}$$

At this time, $[Ag^+]^2[Cr_2O_7^{2-}] < K_{sp}(Ag_2Cr_2O_7)$ and $[Ag^+][OH^-] < K_{sp}(AgOH)$ are also satisfied, which is reasonable.

CHECKPOINT

5 PTS

When Ag_2CrO_4 is just saturated, there is a reasonable solution:

$$[Ag^+] = 3.36 \times 10^{-4}, [H^+] = 9.92 \times 10^{-6}, [Cr_2O_7^{2-}] = 1.06 \times 10^{-5}, [HCrO_4^-] = 3.05 \times 10^{-4}, [CrO_4^{2-}] = 9.94 \times 10^{-6}$$

At this time $pH = 5.00$, option B is wrong.

CHECKPOINT

0.5 PTS

At saturation, $pH = 5.00$

$$[Ag^+]^2[Cr_2O_7^{2-}]/K_{sp}(Ag_2CrO_4) = 1.07, \text{ option C is wrong.}$$

CHECKPOINT

0.5 PTS

$$\text{At saturation, } [Ag^+]^2[Cr_2O_7^{2-}]/K_{sp}(Ag_2CrO_4) = 1.07$$

At saturation, $S = [Ag^+]/2 = 1.68 \times 10^{-4} \text{ mol}$, the dissolved mass $m = S \cdot M_{Ag_2Cr_2O_7} = 7.3 \times 10^{-2} \text{ g}$, option D is wrong.

CHECKPOINT

0.5 PTS

$$S = 1.68 \times 10^{-4} \text{ mol}$$

CHECKPOINT

0.5 PTS

$$m = 7.3 \times 10^{-2} \text{ g}$$

If the two precipitates coexist, then solve Equations 1, 2, 4, 5, and 6 simultaneously to get

$$[Ag^+] = 3.46 \times 10^{-3}, [H^+] = 4.17 \times 10^{-2}, [Cr_2O_7^{2-}] = 1.66 \times 10^{-2}, [HCrO_4^-] = 1.21 \times 10^{-2}, [CrO_4^{2-}] = 9.33 \times 10^{-8}$$

The amount of Cr in the solution $[Cr]_{aq} = 2[Cr_2O_7^{2-}] + [HCrO_4^-] + [CrO_4^{2-}] = 4.53 \times 10^{-2} > 4S$, that is, the amount of dissolved Cr has exceeded the amount added, which is a contradiction. Option E is wrong.

CHECKPOINT

4 PTS

When two precipitates coexist, $[Cr]_{aq} > 4S$

Ag_2CrO_4 is less soluble than $Ag_2Cr_2O_7$, option F is obviously wrong.

CHECKPOINT

0.5 PTS

Ag_2CrO_4 is less soluble than $Ag_2Cr_2O_7$

Therefore, choose A.